Expectation and Midterm 1

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Probability and Statistics for Engineers Fall 2026

Monday, October 12	Expectation, linearity, and indicators
Wednesday, October 14	Review through the binomial distribution
Friday, October 16	Midterm 1

Course support and deadlines

Ed Discussion: questions, announcements, and private messages. No Canvas Inbox.

Open Ed from Canvas All times Pacific Full contact details and meeting links

Instructor / TAs	Office hours	Optional, ungraded discussions
Antonio de Leon	Fri 3–4 p.m. (in person) GSC Room 204	—
Jeannine Romero	Tue / Thu 10–11 a.m.	01A: Fri 9:20–10:25 a.m. R Carson Acad 240 (in person)
Ehsan Mirafzali	Mon 3–5 p.m.	01B: Fri 10:40–11:45 a.m. (Zoom)
Seokjun Choi	Wed 3–4 p.m. 5:05–6:05 p.m.	01C: Wed 4–5:05 p.m. (Zoom)

Upcoming deadlines HW and quizzes: **10 p.m. Pacific**

HW 2	**Mon, Oct 12**	Gradescope
Quiz 3	**Wed, Oct 14**	Canvas
Midterm 1	**Fri, Oct 16**	During class; Classroom Unit 002

Midterm 1 covers required material through Oct 9; expectation is excluded.

HW: 24-hour grace period. Quizzes: no grace period. Approved accommodations apply.



Expectation and the examination boundary

Today: a probability-weighted mean, linearity, and means of counts.

Midterm 1

Required material through October 9. Expectation, variance, and joint distributions are excluded.

Three independent fair coin tosses; X counts heads.

x	0	1	2	3
$p_X(x)$	1/8	3/8	3/8	1/8

Recall: an outcome is a sequence such as HTT. X assigns its head count.

Why are the four possible counts not equally likely?

Expected value: a weighted mean

Finite support $S_X = \{x : p_X(x) > 0\}$

$$\mu = \mathbb{E}[X] = \sum_{x \in S_X} x\, p_X(x).$$

For each value: multiply the value by its probability, then add. The weights satisfy $p_X(x) \geq 0$ and $\sum_{x \in S_X} p_X(x) = 1$. Do not average the support values without their probability weights.

Check: if $a \leq X \leq b$, then $a \leq \mathbb{E}[X] \leq b$. Why?

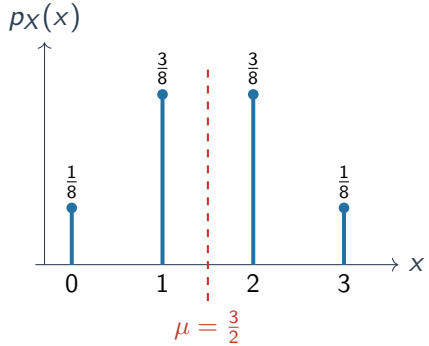
X varies from outcome to outcome; $\mathbb{E}[X]$ is one fixed number.

Finite support today. For countably many values, assume $\sum_x |x| p_X(x) < \infty$.

Mean of a three-toss count

Value $\times$ probability

x	0	1	2	3
$p_X(x)$	1/8	3/8	3/8	1/8



The mean is 1.5 heads, although a single run cannot produce 1.5 heads.

Check: $Y = 0$ with probability $3/4$, or $Y = 2$ with probability $1/4$. Find $\mathbb{E}[Y]$.

A model mean describes the long-run average over independent repetitions, not a guaranteed finite-data average.

Outcomes and values: the same weighted sum

Outcomes	x	$p_X(x)$
TTT	0	1/8
HTT, THT, TTH	1	3/8
HHT, HTH, THH	2	3/8
HHH	3	1/8

Each sequence has probability $1/8$.

$$1\frac{1}{8} + 1\frac{1}{8} + 1\frac{1}{8} = 1\frac{3}{8}.$$

Three outcome contributions become one contribution for the value $x = 1$.

For any finite Ω, write $q_\omega = P(\{\omega\})$. Then

$$p_X(x) = \sum_{\omega : X(\omega) = x} q_\omega, \qquad \underbrace{\sum_{\omega \in \Omega} X(\omega) q_\omega}_{\text{sum over outcomes}} = \underbrace{\sum_{x \in S_X} x\, p_X(x)}_{\text{group by value}}.$$

The outcome probabilities need not be equal. We normally calculate using the support of X.

Linearity of expectation

Finite expectations; constants a, b

$$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y].$$

Why it works on a finite sample space. Evaluate both variables on the same outcome and distribute the weights:

$$\begin{aligned}\mathbb{E}[aX + bY] &= \sum_{\omega\in\Omega} [aX(\omega) + bY(\omega)]q_\omega \\ &= a\sum_{\omega\in\Omega} X(\omega)q_\omega + b\sum_{\omega\in\Omega} Y(\omega)q_\omega \\ &= a\mathbb{E}[X] + b\mathbb{E}[Y].\end{aligned}$$

Independence is not required. No probability was factored into a product.

The last step uses the grouping identity. The outcome formula is useful for this proof; the support formula remains our main calculation rule.

Linearity with dependent variables

In the same three fair tosses, let X count heads and Y count tails.

X	0	1	2	3
Y	3	2	1	0
$X + Y$	3	3	3	3
Probability	1/8	3/8	3/8	1/8

Knowing X determines Y. Nevertheless, compare $\mathbb{E}[X + Y]$ with $\mathbb{E}[X] + \mathbb{E}[Y]$.

Repeated application: $\mathbb{E}[\sum_{i=1}^{n} a_i X_i] = \sum_{i=1}^{n} a_i \mathbb{E}[X_i]$.

Constants and changes of scale

A constant variable $C = c$ has support $\{c\}$, so $\mathbb{E}[C] = c \cdot 1 = c$.

Weighting the transformed values by the original probabilities gives

$$\mathbb{E}[aX + b] = \sum_{x \in S_X} (ax + b) p_X(x) = a\mathbb{E}[X] + b \underbrace{\sum_{x \in S_X} p_X(x)}_{1}.$$

Example. Award $+1$ for a head and -1 for a tail in three tosses. If X counts heads, the score is $Z = X - (3 - X) = 2X - 3$.

Multiplying values by a scales the mean; adding b shifts the mean.

General functions use $\sum_x g(x) p_X(x)$; see the optional reference.

Mean of an indicator

An indicator turns a yes/no event into a numerical random variable:

$$I_A = \begin{cases} 1, & \text{if event } A \text{ occurs,} \\ 0, & \text{otherwise.} \end{cases}$$

Value u	0	1
$p_{I_A}(u)$	$1 - P(A)$	$P(A)$

The same weighted-mean rule gives

$$\boxed{\mathbb{E}[I_A] = 0[1 - P(A)] + 1\,P(A) = P(A).}$$

Check: a request fails with probability 0.02. What is the mean of its failure indicator?

For $I_A \sim \mathrm{Bern}(p)$, $\mathbb{E}[I_A] = p$, also at $p = 0, 1$ (zero-mass terms vanish).

Mean of a count

Let I_i be 1 when trial i is a success and 0 otherwise.

$$N = I_1 + \cdots + I_n.$$

For example, the pattern success, failure, success gives $N = 1 + 0 + 1 = 2$.

$$\mathbb{E}[N] = \sum_{i=1}^{n} \mathbb{E}[I_i] = \sum_{i=1}^{n} P(\text{success on trial } i).$$

When every success probability is p

$$\mathbb{E}[N] = np.$$

If the trials are also mutually independent, $N \sim \mathrm{Bin}(n, p)$.

Check: for the three fair tosses, $\mathbb{E}[X] = 3(1/2) = 3/2$ again.

Independence is needed to justify the binomial model, not this mean calculation.

Sampling without replacement

A bag has three red balls and two blue balls. Draw two uniformly without replacement. Let N count reds; I_i indicates that draw i is red.

1. Express N using indicators and find $\mathbb{E}[N]$.
2. Show that the draws are dependent. Why does your mean calculation still work?

Distinguish the unconditional chance of red from the chance after seeing the first draw.

Expectation: summary

Object or operation	Result
Probability-weighted mean	$\mathbb{E}[X] = \sum_{x \in S_X} x\, p_X(x)$
Change of scale and shift	$\mathbb{E}[aX + b] = a\mathbb{E}[X] + b$
Sum of variables	$\mathbb{E}[\sum_i a_i X_i] = \sum_i a_i \mathbb{E}[X_i]$
Indicator / count	$\mathbb{E}[I_A] = P(A)$; $\mathbb{E}[\sum_i I_{A_i}] = \sum_i P(A_i)$

Explain: can a mean be an impossible observation? Does linearity need independence?

Expectation is excluded from Midterm 1. Functions of X and variance come next.

Midterm 1 review

Scope: events and counting; conditioning, total probability and Bayes; independence; random variables, PMFs/CDFs; Bernoulli and binomial probabilities.

Excluded

Expectation, variance, and joint distributions. Monday's lesson is outside Midterm 1.

We will work selected questions from the released practice paper.

Before each calculation: identify the random experiment, the requested event, and the assumptions.

The practice paper illustrates reasoning and answer detail. It does not predict which topics or question forms will appear on the exam.

Allowed: one page of notes (a cheat sheet) and a basic nonprogrammable calculator.

Concept checks

Released practice: Question 1(a), (b), (d), (f). Decide first; then give a short correction if false.

1. $P(A) = 2/5$ and $P(B) = 3/5$ imply that A and B are disjoint.
2. $P(A \cap B) = P(A)\,P(B)$ implies that A and B are independent.
3. A finite PMF is valid whenever every listed probability is between 0 and 1.
4. Three favorable outcomes out of twelve always give probability $1/4$.

One valid counterexample disproves a universal claim.

Counting codes and conditioning

Released practice: Question 3

Three-letter codes use A, B, C, D, E. Repetition is allowed; all 5^3 codes are equally likely.

(a) Exactly two different letters

How many codes?

(b) Different endpoints

Given this condition, find the probability of exactly two different letters.

A count need not be a probability. A condition changes which outcomes are being counted.

Recovering probabilities from conditional information

Released practice: Question 4(a)–(b)

E: shipped express. D: delivered the next day.

$$P(E) = \frac{2}{5}, \qquad P(D \mid E) = \frac{3}{5}, \qquad P(E \mid D) = \frac{4}{5}.$$

Find $P(D)$ and $P(D \mid E^c)$. Start with the shared intersection.

$P(E \mid D)$ and $P(D \mid E)$ use different denominators.

An event table: exactly one and independence

Released practice: Question 4(c)–(d). The table organizes the probabilities just found.

	D	D^c	Total
E	$12/50$	$8/50$	$20/50$
E^c	$3/50$	$27/50$	$30/50$
Total	$15/50$	$35/50$	1

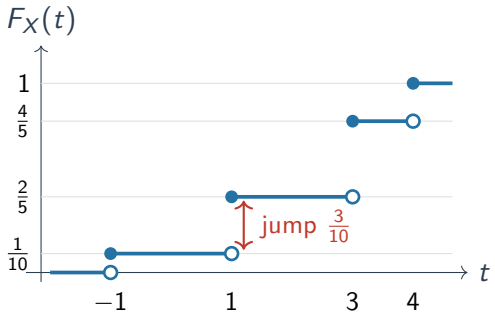
(c) Find the probability that exactly one of E, D occurs.

(d) Are E and D independent? Give a numerical check.

Exactly one excludes the intersection; at least one includes it.

CDF jumps and endpoint probabilities

Released practice: Question 2. This plot is the given piecewise CDF.



(a) Give the support and PMF.
(b) Find $P(X < 1)$ and $P(-1 < X \leq 3)$.

$F_X(t) = P(X \leq t)$. A filled dot gives the value at a jump.

$$p_X(x) = F_X(x) - F_X(x^-), \qquad P(a < X \leq b) = F_X(b) - F_X(a).$$

$F_X(x^-)$: the level just left of x. Here $P(X \leq 1) = 2/5$ but $P(X < 1) = 1/10$.

A binomial model and a tail event

Released practice: Question 5(a)

Six independent sensor readings; each is incorrect with probability $1/5$. Let N count incorrect readings.

State the distribution and support; find $P(N \geq 5)$.

The coefficient chooses the positions of the five incorrect readings.

Model check: fixed number of trials, two outcomes per trial, common probability, mutual independence.

Conditioning on the total number of errors

Released practice: Question 5(b). Keep the six independent readings with error probability $1/5$.

Given $N = 2$, find the probability that the first two readings are both correct.

Position	1	2	3	4	5	6
Required	correct	correct	the two errors must occur here			

Why are the possible pairs of error positions equally likely?

$(4/5)^2$ answers the unconditional question. Conditioning on the total changes the probabilities.

When the binomial model no longer applies

Released practice: Question 5(c)

For this part only, the sixth reading has error probability $1/2$. The first five still have error probability $1/5$; all six are independent.

Find the probability that all six are correct. Is $N \sim \mathrm{Bin}(6, 1/5)$ still valid?

Independence and equal success probabilities are separate assumptions.

Related event: at least one error is the complement of all six correct.

Writing a complete solution

Problem type	What to show
Counting	Explain each choice; show why each outcome is counted once.
Conditional probability	Identify the conditioning event and its denominator.
PMF / CDF	State the support and zero elsewhere; check endpoint inclusion.
Binomial	Give n, p, support, assumptions, and the requested event.
Independence	Use a product or conditional-probability comparison.

Exact expressions may be left unevaluated. Label each part and include the reasoning that leads to your answer.

Use all required course topics when studying. This review is a sample of methods, not an exam blueprint.

Midterm 1

Examination meeting

Friday, October 16, 2026.
55 minutes of working time within the scheduled class period.

Scope

Required material through October 9: events, counting, conditioning, independence, random variables, PMFs/CDFs, and Bernoulli/binomial laws.

Excluded

Expectation, variance, and joint distributions.

Materials

One page of notes (a cheat sheet) and a basic nonprogrammable calculator. No communication or other electronic resources.

Approved accommodations are handled separately. Follow the examination's written instructions.

Optional: expectation of a function

For finite-valued X and a real-valued function g, put $Y = g(X)$. Grouping inputs that give the same output gives

$$\mathbb{E}[Y] = \sum_y y \sum_{x:g(x)=y} p_X(x) = \sum_{x \in S_X} g(x) p_X(x).$$

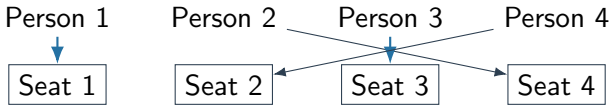
The probabilities stay attached to the original inputs.

x	0	1	2	3
$p_X(x)$	1/8	3/8	3/8	1/8
$g(x) = x^2$	0	1	4	9

For countably supported X, assume $\sum_x |g(x)| p_X(x) < \infty$. This extension is excluded from Midterm 1 and will be revisited before variance.

Optional: random matching

There are $n \geq 1$ labeled seats and n people, each with an assigned seat. Choose one of the $n!$ seating permutations uniformly. Let M count correct matches.

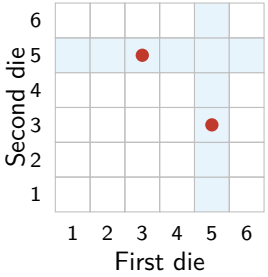


Let I_i indicate that person i gets seat i. Find $\mathbb{E}[M]$.

The diagram shows one seating of four people. The mean calculation covers all permutations.

Optional: changed conditioning

Roll two independent fair dice. Let S be their sum and B the event that at least one die shows 5.



Compare three observations

1. At least one die shows 5.
2. The first die shows 5.
3. Exactly one die shows 5.

What is the probability of $S = 8$ under each observation?

Count favorable outcomes inside the conditioning event.

References for this week

DeGroot and Schervish

Probability and Statistics, fourth edition.

- §4.1, pp. 207–209: discrete expectation and Bernoulli example.
- §4.2, pp. 217–221: affine changes, linearity, and sampling means.

Mor Harchol-Balter

Introduction to Probability for Computing, supplied 2024 version.

- §4.1, pp. 58–61: expectation and functions of a discrete variable.
- §4.2, pp. 63–64: linearity and the binomial mean.
- Example 4.12, pp. 64–65: optional random matching.

Selections only. Review: current Weeks 1–2 slides and the released Midterm 1 Practice and Solutions (October 8 revision).

Additional source: Sheng Jiang, STAT 131 Notes (July 18, 2024), pp. 48–51; Antonio de Leon's earlier STAT 131 materials.